

9. On the critical parabola

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On the critical parabola, $\det \mathbf{A} = (\text{tr}^2) \mathbf{A} / 4$, the systems have a repeated real eigenvalue. When the eigenvalue is repeated, there are two possibilities for the space of eigenvectors. The space of all eigenvectors for this only eigenvalue may be 1- or 2-dimensional. Depending on whether the repeated eigenvalue is nonzero or zero, and whether the space of eigenvectors is 1- or 2- dimensional, the phase portraits are different, as follows.

Suppose that there is a **repeated real eigenvalue**, $\lambda_1 = \lambda$, $\lambda_2 = \lambda$. The space of eigenvectors of λ could be either 1-dimensional or 2-dimensional.

1. **Degenerate node** (or **improper node** or **defective node**):
Repeated nonzero eigenvalue with **1-dimensional** space of eigenvectors, $\lambda \neq 0$ and $\mathbf{A} \neq \lambda \mathbf{I}$.

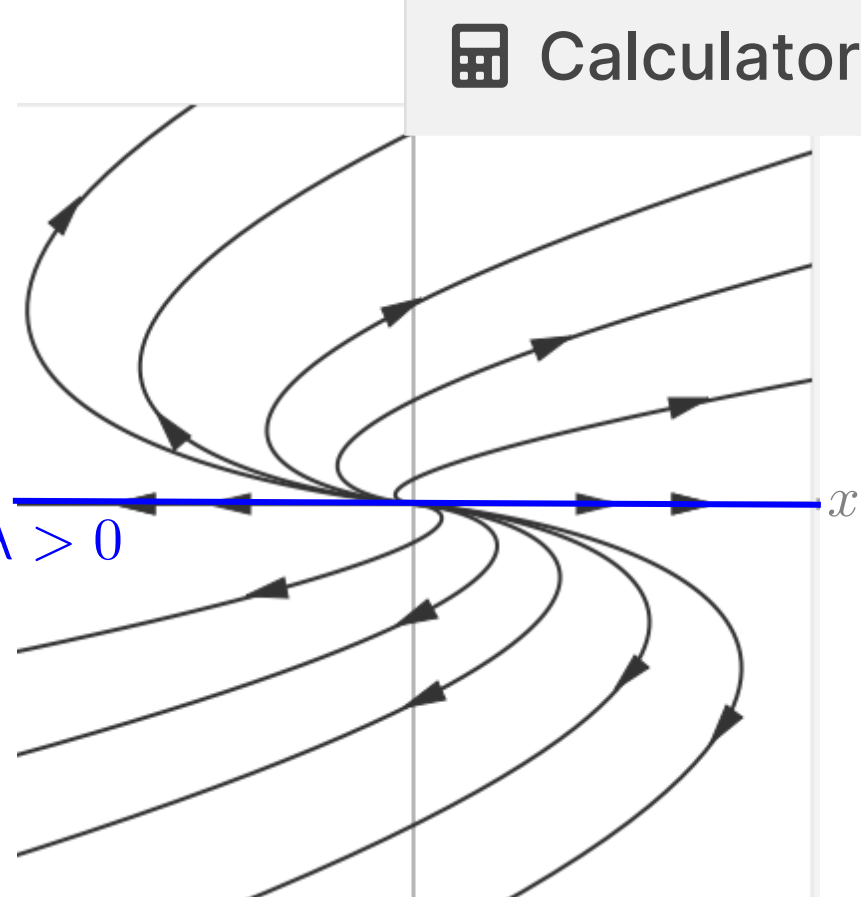
There is just one eigenline.
Every trajectory not contained in the eigenline becomes tangent to it at $(0, 0)$, and becomes parallel to it when far away from $(0, 0)$. Such trajectories are

- repelled from $(0, 0)$ if $\lambda > 0$,
- attracted to $(0, 0)$ if $\lambda < 0$.

This is a borderline case between a node and a spiral.

Example of solution:

$$\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + c_2 \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} e^t + \begin{pmatrix} 2 \\ 0 \end{pmatrix} t e^t \right)$$



Calculator

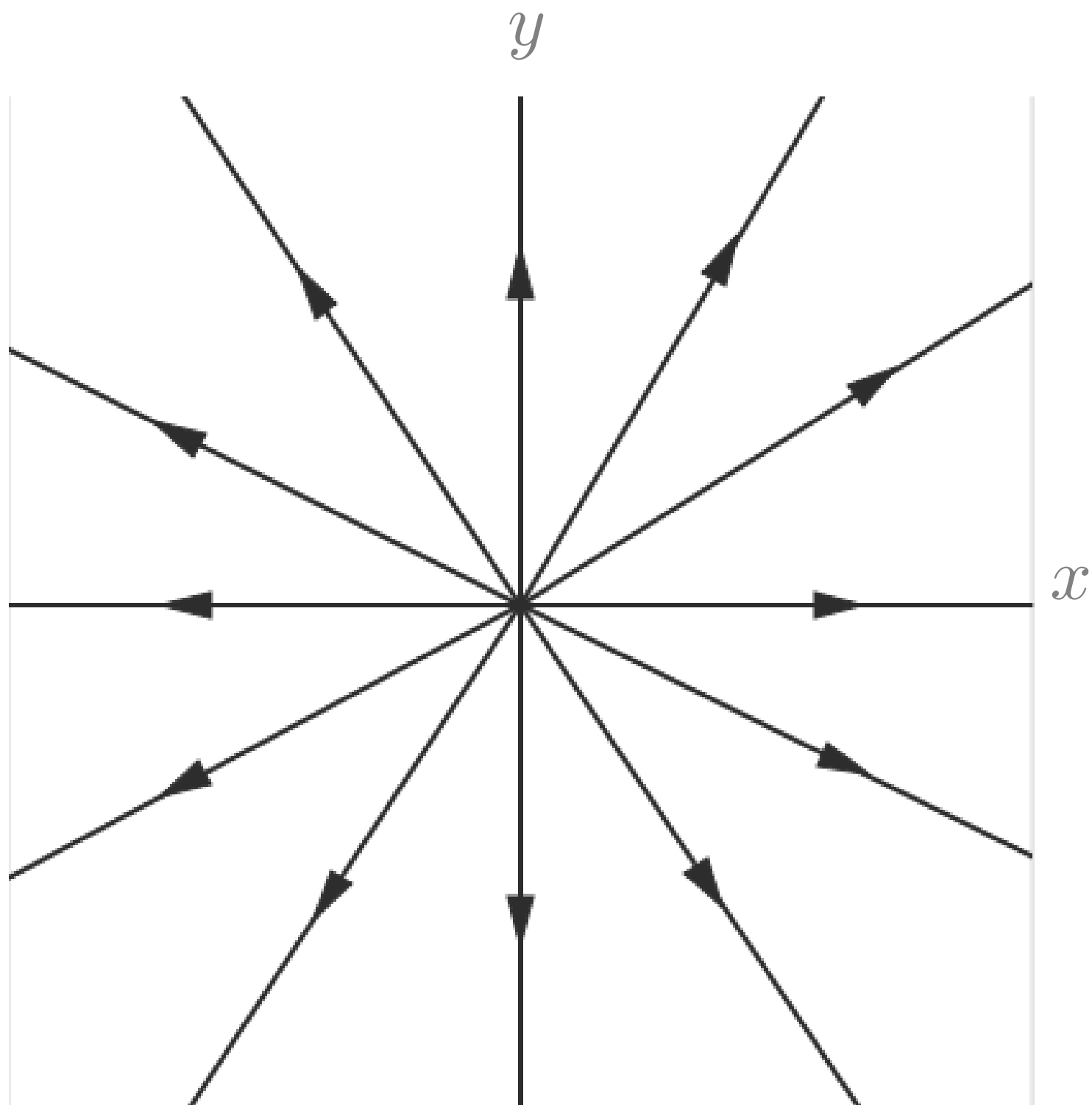
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2. **Star Node:**
Repeated nonzero eigenvalue, 2-dimensional space of eigenvectors, $\lambda \neq 0$ and $\mathbf{A} = \lambda \mathbf{I}$.

Every nonzero vector is an eigenvector.
Nonzero trajectories are along rays, and are

- repelled from $(0, 0)$ if $\lambda > 0$,
- attracted to $(0, 0)$ if $\lambda < 0$.

General solution: $\mathbf{x} = e^{\lambda t} \left[c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]$.



At the origin of the trace-determinant plane, both eigenvalues are zero. There are two kinds of phase portraits, depending on the dimension of the space of eigenvectors.

1. **Repeated zero eigenvalue** with **1-dimensional** space of eigenvectors, $\lambda = 0$ and $\mathbf{A} \neq \mathbf{0}$. There is only one eigenline.
Points on the eigenline are stationary. All other trajectories are lines parallel to the eigenline.

General solution: $\mathbf{x} = c_1 \mathbf{v} + c_2 (t \mathbf{v} + \mathbf{w})$ where $\mathbf{v}$ is an eigenvector of $\mathbf{0}$ and the constant vector $\mathbf{w}$ satisfies $\mathbf{v} = (\mathbf{A} - \lambda \mathbf{I}) \mathbf{w}$.

2. **Repeated zero eigenvalue** with **2-dimensional** space of eigenvectors, $\lambda = 0$ and $\mathbf{A} = \mathbf{0}$.

Every solution is a **stationary point**.

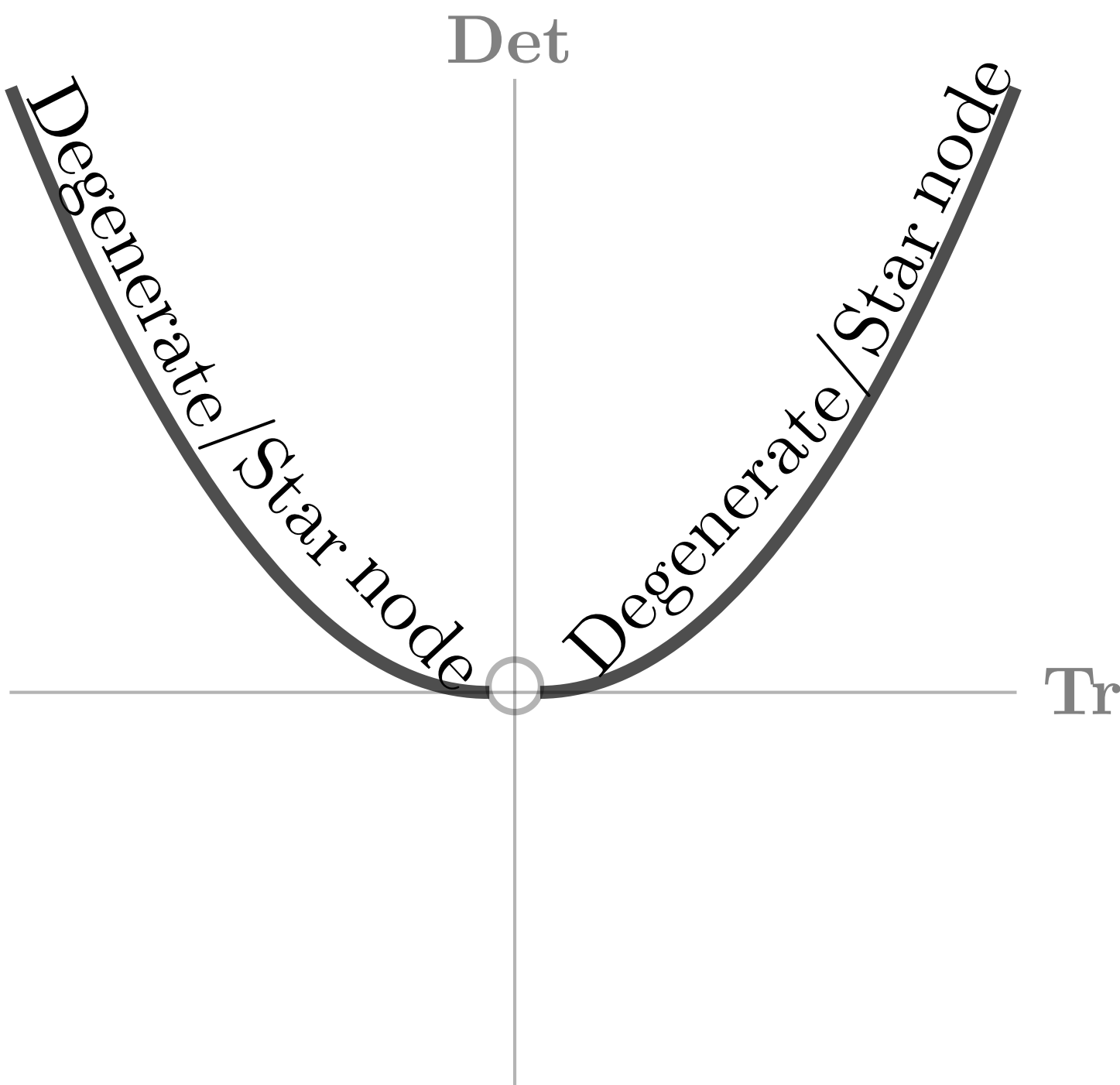
General solution: $\mathbf{x} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$.

On the trace-determinant plane:

Suppose that $\det \mathbf{A} = (\text{tr} \mathbf{A})^2 / 4$.

Then we get a repeated real eigenvalue $\lambda_1 = \lambda$, $\lambda_2 = \lambda$, so that $\text{tr} \mathbf{A} = 2\lambda$.

- If $\text{tr} \mathbf{A} \neq 0$, degenerate node or star node.
- If $\text{tr} \mathbf{A} = 0$, stationary solutions or solution along lines parallel to eigenline.



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Degenerate node or star node

discussion posted 3 months ago by Marco_Str

I don't understand the two cases very well: where do the solutions come from? It would be good to calculate an example in both cases.

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1 response

MarkvdV

3 months ago - endorsed 3 months ago by jfrench (Staff)

Hi, there is a practice in recitation 4 of this unit regarding this topic, so that's maybe why it is not given here. The way to get to the solution is to inspect the system of DE's and recognize a standard way to solve the DE's, without having to determine eigenvalues and eigenvectors. Following that way and rearranging the notation of the general solutions will lead to the form of the example solution for a degenerate node given here.

I was thinking the same thing

posted 23 days ago by Qasim_321

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